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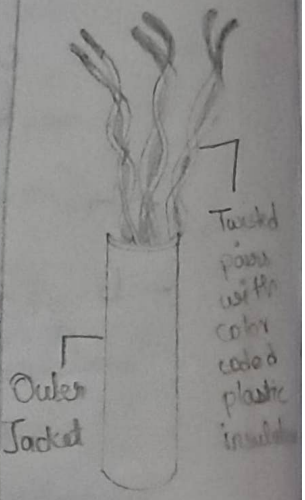
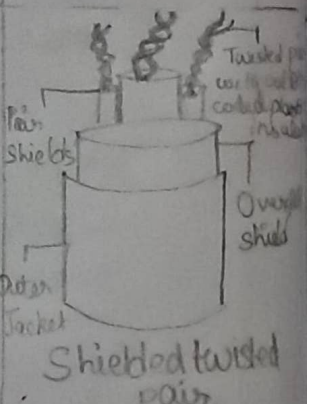
Practical - 2

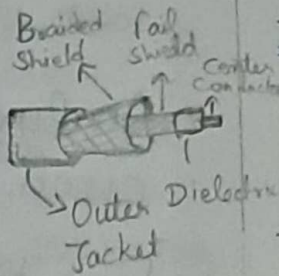
Aim: Study of different types of Network cables

a) Understand different types of network cables

Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/ Use	Image
UTP	Category 3	10 bps	Advantages: • Cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantages: • More prone to (EMI) Electromagnetic Interference and Noise	10 Base-T Ethernet	
	Category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a	10 Gbps	Advantages • Shielded • Faster than UTP • Less susceptible to noise and interference	Gigabit Ethernet, 10G Ethernet. (55 m) Widely used in data centers	
SSTP	Category 7	10 Gbps	Disadvantages: • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (100 m)	

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: • Higher bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadvantages: • Limited distance • Cost • Size is bulky	Speed of signal is 500 m Television network High speed internet connections	
Fibre Optics Cable	Single Mode Multi Mode	100 Gbps	Advantages: • High Speed • High Bandwidth • High Security • Long Distance Disadvantages • Expensive • Requires skilled installers	Maximum distance of fibre optics cable is around 100 kms	

Result:
Thus different types of cables are studied successfully.